

AHComputing Roofing Estimates

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ROOFING
ESTIMATE

BILL TO
Property Owner
3561 E 102nd Ct
Thornton, CO 80229

Estimate # RE-202605-6497
Estimate Date 05-08-2026
Valid Until 05-22-2026

Roof Area 2081 sqft	Squares 20.8	Eave Length 133 Inft	Ridge/Hip 26 Inft	Roof Type Hip	Pitch steep (9/12)	AI Confidence High
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QTY	UNIT	DESCRIPTION	UNIT PRICE	AMOUNT
71	bundles	Architectural Roofing Shingles	\$48.00	\$3,408.00
6	rolls	Roofing Underlayment (15lb felt, 400 sqft/roll)	\$75.00	\$450.00
399	sqft	Ice & Water Shield (eave protection)	\$0.75	\$299.25
133	Inft	Drip Edge / Eave Trim	\$1.50	\$199.50
26	Inft	Ridge Cap Shingles (ridge + hips)	\$3.50	\$91.00
3	Inft	Valley Flashing	\$4.00	\$12.00
53	lbs	Roofing Nails	\$3.50	\$185.50
20.8	squares	Tear-Off & Disposal (existing roof removal)	\$65.00	\$1,352.00
20.8	squares	Installation Labor	\$150.00	\$3,120.00

Subtotal \$9,117.25

Sales Tax (8%) \$729.38

TOTAL (USD)	\$9,846.63
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Estimator Notes: Solar API data: 8 roof segments, avg pitch 36.3° (9/12), imagery from 2024-07. Quality: HIGH.

Terms & Conditions: This estimate is valid for 30 days. A 50% deposit is required to begin work. Final invoice may vary +/-10% based on field conditions. All work performed to local building code. Permit costs not included.

Contractor Signature: _____

Customer Signature: _____

Roof Analysis & Calculation Detail

Estimate RE-202605-6497 | 3561 E 102nd Ct, Thornton, CO 80229

SATELLITE IMAGE



Zoom 20 | 0.0573 m/px
640x640px | Google Maps Static API
Lat 39.88116

ROOF PLANE BREAKDOWN

PLANE	FOOTPRINT	PITCH	MULT	ADJ AREA
West-facing slope	554 sqft	9/12	x1.2529	694.1 sqft
East-facing slope	309 sqft	8/12	x1.2105	374.0 sqft
East-facing slope	220 sqft	8/12	x1.209	266.0 sqft
East-facing slope	243 sqft	11/12	x1.336	324.6 sqft
North-facing slope	189 sqft	8/12	x1.2114	229.0 sqft
South-facing slope	165 sqft	8/12	x1.2072	199.2 sqft
South-facing slope	161 sqft	9/12	x1.2555	202.1 sqft
West-facing slope	147 sqft	9/12	x1.2389	182.1 sqft

Footprint total: 1,988 sqft | Avg multiplier: 1.047 | Adjusted area: 2,081 sqft

Calculation Detail — Show Your Work

ITEM	FORMULA
Adjusted area (waste)	$2,081 \text{ sqft} \times (1 + 12\% \text{ waste}) = 2,330.7 \text{ sqft}$
Shingle bundles	$2,330.7 \text{ sqft} / 33 \text{ sqft/bundle} = 70.6 \rightarrow 71 \text{ bundles (ceiling)}$
Underlayment rolls	$2,330.7 \text{ sqft} / 400 \text{ sqft/roll} = 5.83 \rightarrow 6 \text{ rolls (ceiling)}$
Ice & water shield	$\text{Eave } 133 \text{ lnft} \times 3 \text{ ft protection band} = 399 \text{ sqft}$
Drip edge	$\text{Eave perimeter} = 133 \text{ lnft}$
Ridge cap	$\text{Ridge} + \text{hips} = 26 \text{ lnft}$
Valley flashing	$\text{Valley length} = 3 \text{ lnft}$
Roofing nails	$20.8 \text{ squares} \times 2.5 \text{ lbs/sq} = 52.0 \rightarrow 52 \text{ lbs (ceiling)}$
Labor (squares)	$2,081 \text{ sqft} / 100 = 20.8 \text{ squares}$
Sales tax	$\text{Subtotal } \$9,117.25 \times 8\% = \729.38

Google Solar API — Measurement Method

Roof measurements provided by the Google Solar API (buildingInsights endpoint). Google's Solar API uses high-resolution aerial imagery and machine learning to map individual roof segments with measured pitch, area, and orientation. Data covers 8 roof segment(s). Imagery date: 2024-07. Imagery quality: HIGH. Total roof area 2,081 sqft and pitch data are directly measured values, not estimates — no shadow inference required.

Note: Solar API data reflects the most recent available aerial imagery for this property. Linear measurements (eave, ridge, valley) are estimated from segment geometry. A licensed contractor should verify quantities during physical inspection.